

Table 1. Predictive model performance on the held-out time-based test period.

Model	Accuracy	Precision (Class 1)	Recall (Class 1)	F1 Score (Class 1)	Weighted F1	ROC-AUC
Logistic Regression	0.574	0.129	0.810	0.222	0.670	0.727
Decision Tree	0.899	0.391	0.605	0.475	0.909	0.728
Random Forest	0.899	0.391	0.605	0.475	0.909	0.857

Baseline reference. Perfect agreement is expected because the Week-4 rule was used to define the target label.

Approach	Accuracy	Precision (Class 1)	Recall (Class 1)	F1 Score (Class 1)	Weighted F1	ROC-AUC
Week-4 rule reference	1.000	1.000	1.000	1.000	1.000	1.000

Note. Bold values indicate the single best-performing predictive model for each metric based on the **unrounded** scores; values are displayed to three decimal places. Therefore, values that appear tied after rounding are not both bolded when their underlying unrounded scores differ. Precision, Recall, and F1 Score refer to Class 1 (refresh-candidate class). The Week-4 rule reference is shown separately because it defines the target label itself; its perfect scores represent label-definition agreement rather than independent predictive performance.